

A01 – Report

Windows Network Programming

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Thursday Lab

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Requirements

1. The application must run on multiple computers.
 - a. One computer will act as the server as windows service
 - b. At least one other computer will act as a console client and can be the source of multiple console client threads
2. The server must receive data from concurrent clients and write it to a local file. The server file write method must be a task to allow asynchronously writes to the output file. The file must contain the incoming data sent from the client threads.
3. When the file reaches a certain size, it must notify the clients so they can stop sending data. It must then stop all local tasks. These stops must be graceful.
4. Every time the client sends a message, the server checks if the file size has been reached. If file size is reached, server will notify the client that the max file size has been reached, and the client will stop sending messages.
5. Measure performance on the server: number of clients it takes for performance to drop, max file size, buffer size, and size of the message
6. Server will write the metrics to a log file for proper documentation.
7. Make the IP Address and port manageable for the client
8. Make the IP Address, port, file size, output file name, and logger file name manageable for the server

Observations

Note: The tests are run until the max file size reached.

Number of Clients

Note: The buffer size (4096B), message size (128B), and file size (1,000,000B/1MB) are constants for this category of tests.

50 Clients

The total write time to max file size given the constants was **23198ms**.

100 Clients

The total write time to max file size given the constants was **22555ms**.

250 Clients

The total write time to max file size given the constants and was **23946ms**.

500 Clients

The total write time to max file size given the constants and was **25603ms**.

Buffer Size

Note: The number of clients (100), message size (128B), and file size (1,000,000B/1MB) are constants for this category of tests.

2048 bytes

The total write time to max file size given the constants was **24879ms**.

4096 bytes

The total write time to max file size given the constants was **23565ms**.

8192 bytes

The total write time to max file size given the constants was **23713ms**.

Message Size

Note: The number of clients (100), buffer size (4096B), and file size (1,000,000B/1MB) are constants for this category of tests.

128 bytes

The total write time to max file size given the constants was **24631ms**.

256 bytes

The total write time to max file size given the constants was **12525ms**.

512 bytes

The total write time to max file size given the constants was **6204ms**.

1024 bytes

The total write time to max file size given the constants was **3248ms**.

20000 bytes

The total write time to max file size given the constants was **1268ms**.

Conclusion

Number of Clients

We expected the performance to increase as client number increased but contrary to the results we found, the evidence pointed towards a dip in performance between the 100-client mark and 250-client mark.

Buffer Size

The performance was expected to increase as buffer size increased, but as observed the time to write to file size limit increased by a slight amount when tested with 8192 bytes.

Message Size

We expect decreased performance as the message size increased. The actual results showed an exponential relationship between the completion time and message size.

In summary, the performance testing revealed that system behavior did not always align with the initial expectations. these test cases deepened our understanding of relationships between different variables when it comes to performance testing.